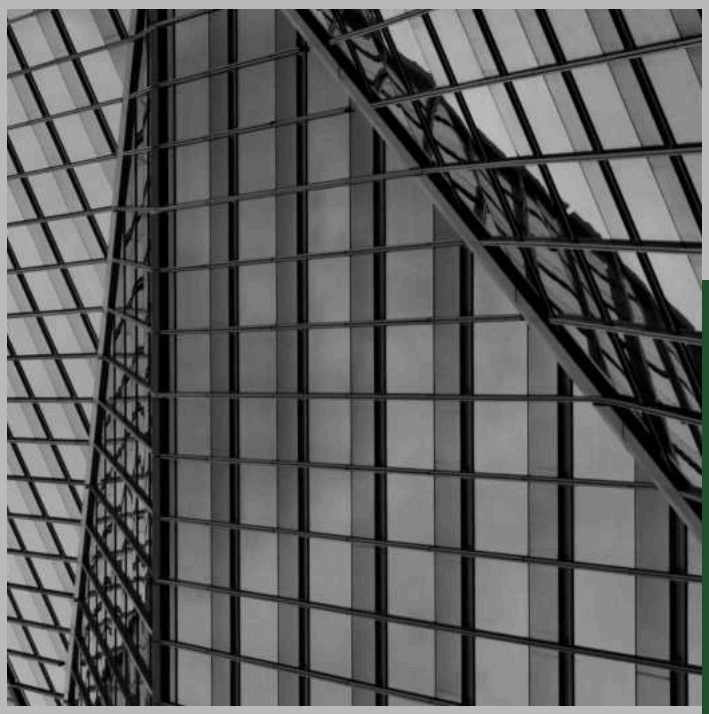
ENGINEERING PORTFOLIO



14023

SPRINGBANK ROBOTICS TEAM

RESURRECTION

THE TEAM

MISSION STATEMENT

> We inspire innovation in robotics, empowering our team and community through STEM education and collaboration, fostering future leaders in technology.

Springbank Robotics Team (SRT) Resurrection is an 8th-year FTC team in Calgary, Alberta.

Our student-led robotics team promotes collaboration, with experienced members mentoring newcomers. We develop leadership and resilience through challenges, gaining essential life skills for future success. Thanks to our mentor, Mr. Mathew, for guiding us into future STEM leaders.

Tyce
Team Lead

Kale
Team lead

Thomas
Build/Drive

Tanner
Build/Coach

Siera
Programming

Callista
Outreach

Hunter
Build

Arianna
Outreach

Nick
Jr Member

Callum
Jr Member

Spencer*
Jr Member

Carter
Outreach

Angil
Jr Member

14023
Robot

OUTREACH AND FUNDRAISING GOALS

- [x] Recruit new members
- [x] Spread awareness of SRT and FIRST
- [x] Raise $10,000 through fundraising
- [ ] _____

OUR GOALS

We made our goals by holding meetings dedicated to discussing what we want to achieve and improve within SRT and writing them down.

To achieve these goals, we took every opportunity we could to reach out to help the community and bring more recognition to SRT!

Some things we did this season!

We donated food to the Radiance Family Society, for families experiencing domestic violence.

We hosted a toy drive with the help of Castle Toys to children in need for the Christmas spirit.

We volunteered at Brenda's House, helping homeless families which is run by the Children's Cottage Society.

We took a trip to the Ecole Valley Elementary School to inspire the 4th graders about STEM and FTC!

MORE OUTREACH!

SRT recently hosted an event for kids and families, offering a hands-on experience in robotics. Guests toured the Springbank Innovation Lab and had a close-up to the team's robots. Robot-building kits were provided, and our guests and team members created their own robots. An obstacle course was also set up for guests to drive student-built robots. The event successfully inspired young students and their families about STEM and robotics.

INSPIRING FUTURE GENERATIONS

SRT is focused on both succeeding in FTC and starting an FLL team with our sister school, Ecole Valley Elementary. We recently presented to 4th graders, showcasing our robot designs and introducing them to the STEM field.

SRT aims to inspire the next generation of innovators in ICT and we believe it's essential to pass down knowledge to younger generations to ensure the program's continued success.

A FRESH START, LIKE A PHOENIX RISING!

Over the past FTC seasons, SRT used separate team names and simple black shirts with a green phoenix. Inspired by other teams' unique and recognizable outfits, we decided to rebrand using names tied to our school's phoenix mascot and its cycles: Fire, Genesis, and Resurrection. As seniors, we chose Resurrection to symbolize our final high school stage and growth. This rebranding also included introducing distinct tracksuit colors for each sub-team, improving recognition and helping us stand out at events. The change reflects SRT's commitment to unity and rising stronger as a team.

OUTREACH

Outreach is crucial for spreading awareness about FTC and inspiring the next generation of innovators. By engaging with schools, businesses, and the community, SRT can show others the excitement of robotics and the opportunities within STEM. Outreach helps build support for the program, fosters valuable leadership skills, and encourages others to get involved. Through outreach, our team not only grows but also contributes to the broader mission of inspiring and empowering future engineers, scientists, and problem-solvers.

TOTAL OUTREACH HOURS			
DATE	**EVENT**	**HOURS**	**IMPACT**
11/24/2024	Presentation at Elbow Valley Elementary School about FTC and SRT	1.5	110 people
12/07/2024	Toy Drive - During Meet #2 at Springbank Community Highschool	6	150 people
12/23/2024	Volunteering at Brenda's House	4	20 families
1/20/2025 to 1/23/2025	Supporting Coding week at Elbow Valley Elementary School	18	200 people
2/4/2025	SHAD Ocean Presentation	1	40 people
3/4/2025	Family and Friends Event at SCHS	2.5	60 people

FUNDRAISING EFFORTS:

09/19/2024
Tridyne Projects Sponsorship
+$10,000.00

10/13/2024 - 12/1/2024
Purdy's Holiday Fundraiser
+$2,696.53

12/07/2024
Journey Engineering Sponsorship
+$2,500.00

1/17/2025
Team Bottle drive
+$618.55

...AND MANY MORE!

SUSTAINABILITY

Our team financial sustainability relies on a mix of sponsor support and our own fundraising efforts. Sponsors are a key part of our success, providing funding that is strictly allocated to cover the costs of building the robots and attending competitions. Their support allows us to focus on the technical aspects of the team without worrying about competition fees and materials.

For other expenses, like our recent rebranding efforts, we raise our own money through fundraisers. For example, our Purdy's chocolate fundraiser had a sales goal of $7,000, and we exceeded it by over $3,000, reaching $10,272.50 in sales. This helped us generate a profit of $2,696.53, which we can use for things like uniforms, outreach activities, and other team needs. Plus, we received a bonus cheque for selling over $10,000, which was a great reward for the hard work of our team. By balancing sponsor contributions and our own fundraising, we ensure that we're financially sustainable and able to reinvest in both our technical and team-building goals.

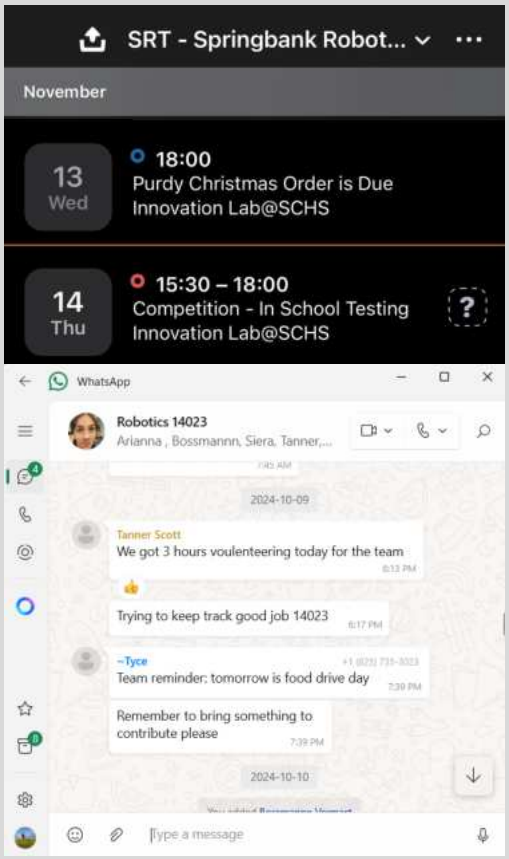
TEAM PLAN & MANAGEMENT

Our FTC robotics team is lucky to have amazing mentors like Mr. Mathew, Heidi, and Trac, who guide us through everything from building and programming to strategy and teamwork. Their support is crucial, but at the heart of our team is the fact that SRT is student-led. This structure empowers us to take ownership of our roles and make decisions that shape the team. Being student-led helps us develop important skills like leadership, problem-solving, and collaboration, while also giving us the freedom to think creatively and learn from our mistakes. With our mentors offering guidance and us driving the direction, we strike the perfect balance between support and independence.

Mr. Mathew
Mentor

Trac Bo
Mentor

Heidi
Mentor



To stay connected and keep everything running smoothly, we use WhatsApp for quick communication between team members. It allows us to share updates, ask questions, and stay on top of any last-minute changes. For scheduling and organizing all of our SRT teams, we rely on TeamSnap. It keeps everyone in sync with practice times, meetings, and competitions, ensuring we're all on the same page and can plan ahead. These tools make collaboration more efficient and help us stay organized, so we can focus on what really matters, building and improving our robots.

DESIGN

MEET 14023'S ROBOT

ROBOT DESIGN FEATURES

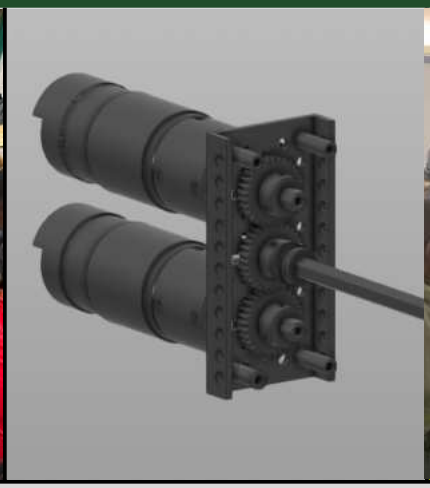
We started off the year knowing that we wanted to do an active intake design. We saw potential in quickly picking up samples and then placing them in the baskets during the matches to gain more points than if we were to hang specimens. With this in mind, we knew our robot had to be relatively big in size to ensure stability, as well as strong parts to avoid breaking down mid-match.

CURRENT PROBLEMS	DESIGN SOLUTION
Viper slides were still slow compared to the drive motors, and added additional weight which led to speed and hanging issues	Upgraded equipment opting for a new supplier, Misumi. The new material was lighter and increased the speed without compromising integrity and strength
The new viper slides were shorter and caused issues with both reaching the height of the basket and being able to reach the blocks on the field	Created new towers to add additional height and space for an additional viper slide and CAD'ed a new horizontal extension bracket

BRAINSTORMING & SKETCHES

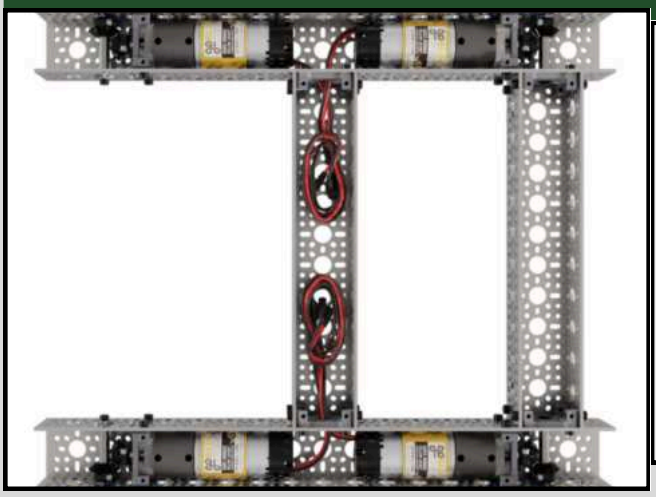
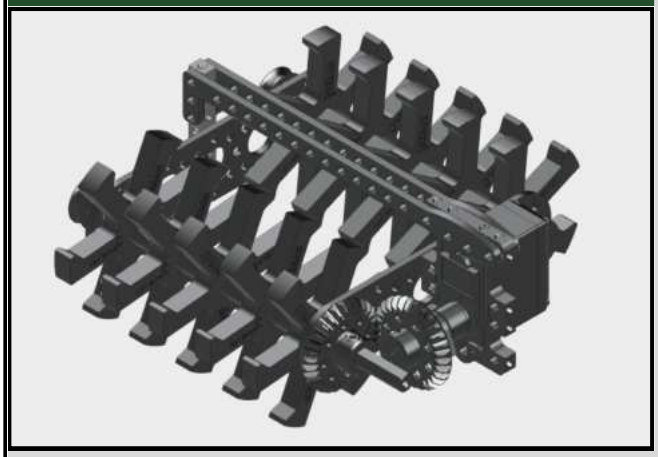
The day FTC launched the game, our team went to work right away, and began brainstorming ideas. Then, later that day in a meeting with the other two SRT teams, we discussed the strategies and different design ideas we came up with, and helped each other find pros and cons of each. Then, we drew our ideas out on paper to help build on what we discussed, and off we went!

BRAINSTORMING



After deciding to try several ideas, we moved on to the decision stage. We narrowed our ideas and creations, one after the other in order, to decide what elements we wanted on our robot. We then drew a rough draft of what the robot would look like on a whiteboard so that we could envision our creation.

CAD

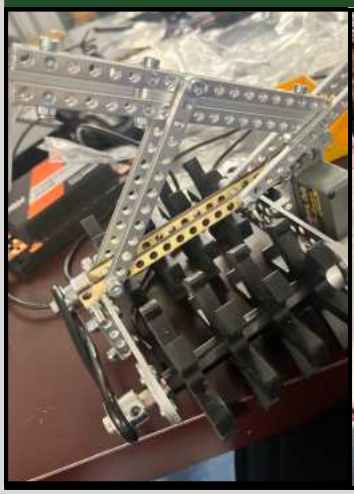
PROTOTYPE AND ASSEMBLY

Once our robot sketches were complete, we then moved on to the prototype stage. We made a 3D model of our robot (CAD) on Autodesk Fusion360 to make sure parts would fit properly, and once that was complete, we moved on to assembling the robot!

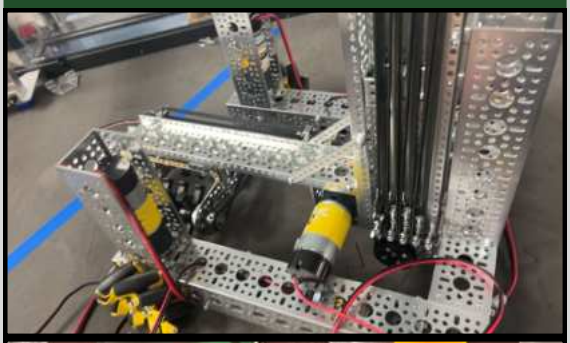
CLAW PROTOTYPE

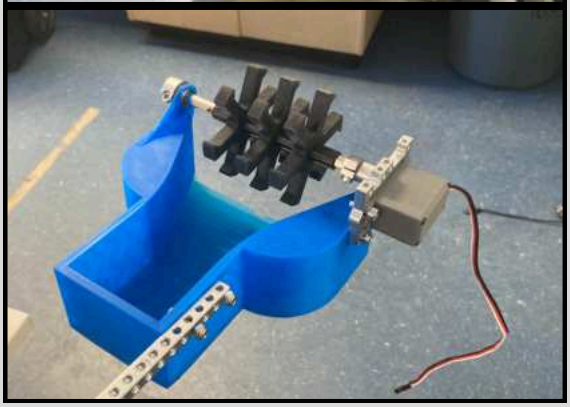
After the third competition, we decided to explore a claw design with an active intake and angled ramp. Unfortunately, after extensive testing, we found it to be much less efficient than our old one, prompting us to return to the system using two servos and intake wheels on axles. This change showed our ability to learn, adapt, and balance performance with creativity. By embracing innovation, we keep improving and refining our ideas for better results.

ASSEMBLY



PROTOTYPES





During the assembly stage, we took parts of the prototype which we were satisfied with. If we weren't happy with our prototype, we'd repeat the design stages to get back to this stage. We then completed our CAD to see the parts we needed to buy, as well as specialized parts we needed to design.

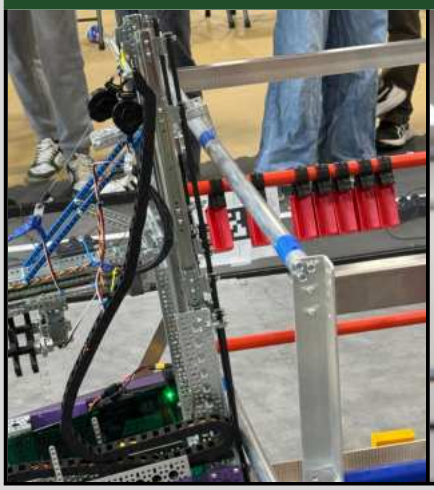
EVALUATE AND ITERATE

Once the robot was assembled, it was ready for evaluation. As such, we tested our robot on the field and saw how it ran. After evaluation, we had a meeting discussing the strengths and weaknesses of our robot. We then sorted out what parts of the robot we want to keep and the parts we could improve on.

EVALUATION

Through multiple iterations targeted at our weaknesses, we continuously worked towards making our robot more reliable, and consistent in every aspect. This took several attempts, but it was worth it when we reached our final robot and completed our CAD!

ITERATION



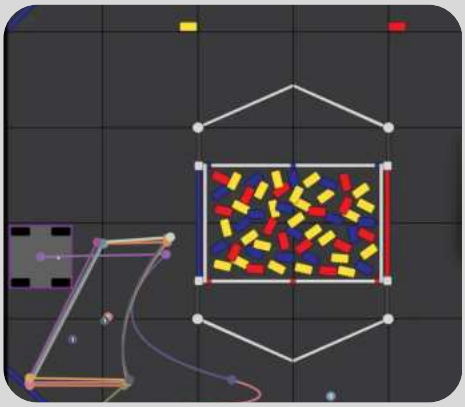
...THEN WE REPEAT!

AUTONOMOUS

CONTROL

ROBOT AUTONOMOUS CODE

This year, we prioritized autonomous for its impact on our success. We coded it using Java and utilized dead wheels for precision. Initially, we used Roadrunner but switched to PedroPathing for its self-correction feature when off course, improving consistency and efficiency.



A NOTE ON PEDRO PATHING

Designed by team 10158, PedroPathing is an advanced Reactive Vector Follower. Essentially, it means our robot can adjust to obstacles or events that happen during its path in real time. New and largely un-tested, for us, Pedro has shown itself to be more reliable, faster, and accurate than the traditional follower RoadRunner. Although it is more difficult to set up and much more time consuming, it will significantly increase our autonomous reliability and consistency.

PROBLEMS	PROGRAM SOLUTION
With Roadrunner, we found that coding the pathing and trajectories was often difficult and tedious, often consuming a significant amount of time.	PedroPathing has a visualizer created to help map out the pathing we wanted the robot to follow.
Roadrunner can only correct its mistakes after the path that it is running finishes, leading to unreliability and inconsistency.	PedroPathing can correct during the path it is running. It adapts to events and obstacles in real time, making it significantly more reliable.

TELE-OP

ROBOT TELE-OP CODE

Similarly to autonomous, we wanted the Tele-Op code to be as efficient and effective as possible. This is why we switched our generic field centric code to a PedroPathing based field centric. This allowed for smoother and more efficient driving, maximizing our scoring potential.

```java
package org.firstinspires.ftc.teamcode.drivecode;

import com.pedropathing.follower.Follower;
import com.pedropathing.localization.Pose;
import com.pedropathing.util.Constants;
import  com.qualcomm.robotcore.eventloop.opmode.OpMode;
import com.qualcomm.robotcore.eventloop.opmode.TeleOp;
import com.qualcomm.robotcore.hardware.CRServo;
import com.qualcomm.robotcore.hardware.DcMotor;
import com.qualcomm.robotcore.hardware.DcMotorSimple;

import org.firstinspires.ftc.teamcode.pedroPathing.constants.FConstants;
import org.firstinspires.ftc.teamcode.pedroPathing.constants.LConstants;

Rename usages
@TeleOp(name = "14023PedroFieldCentricDrive", group = "Linear OpMode")
public class PedroFieldCentric extends OpMode {
    8 usages
    private Follower follower;
    1 usage
    private final Pose startPose = new Pose( setX: 0, setY: 0, setHeading: 0);


    2 usages
    private DcMotor front_left;
    3 usages
    private DcMotor front_right;
    2 usages
    private DcMotor back_left;
    3 usages
    private DcMotor back_right;
```

PROBLEMS	PROGRAM SOLUTION
Robot-oriented code was not the best option for our drivers, as they kept getting disoriented during play.	We changed the robot-oriented code to Pedropathing field-centric code, utilizing the deadwheels on our robot to give our drivers better control.
The lift system on the robot kept falling, as the weight of the intake system was too much for the motors to handle without power.	We changed the default power of the motors from 0 to 0.05, in order to keep the lift where it was when the controller wasn't being pressed.
Due to the height of our fully extended lift system and the weight of the extra motors, our robot became extremely unstable when doing micro-adjustments in and around the basket.	We decided to add a "speed button" within our code to mitigate the risk of the robot tipping over when placing samples in the baskets. It reduced the sudden movements, making small adjustments easier.

MODIFICATIONS

We are always looking for ways to improve, and we know that as we continue in the season our competition grows stronger. We are continuously finding news ways to make our robot better than the last competition. Our adjustments aren't just to keep up with competitors, but to exceed our personal best when it comes to our skills that are reflected in our robot!

CURRENT ROBOT

For provincials, our goal was to keep our fundamental robot the same while making key tweaks to improve efficiency. This included upgrading our vertical and horizontal slides and improving our autonomous to enhance accuracy. These changes helped us optimize performance while maintaining the integrity of our original design.

```
grabPickup1 = follower.pathBuilder()
        .addPath(new BezierCurve(new Point(scorePose), new Point(firstControlPose), new Poi
        .setLinearHeadingInterpolation(scorePose.getHeading(), pickup1Pose.getHeading())
        .build();

/* This is our scorePickup1 PathChain. We are using a single path with a BezierLine, which
scorePickup1 = follower.pathBuilder()
        .addPath(new BezierCurve(new Point(pickup1Pose), new Point(firstControlPose), new P
        .setLinearHeadingInterpolation(pickup1Pose.getHeading(), scorePose.getHeading())
        .build();

/* This is our grabPickup2 PathChain. We are using a single path with a BezierLine, which i
grabPickup2 = follower.pathBuilder()
        .addPath(new BezierCurve(new Point(scorePose),new Point(secondControlPose), new Poi
        .setLinearHeadingInterpolation(scorePose.getHeading(), pickup2Pose.getHeading())
        .build();

/* This is our scorePickup2 PathChain. We are using a single path with a BezierLine, which
scorePickup2 = follower.pathBuilder()
        .addPath(new BezierCurve(new Point(pickup2Pose),new Point(secondControlPose),new Po
        .setLinearHeadingInterpolation(pickup2Pose.getHeading(), scorePose.getHeading())
```

For our autonomous, we wanted to try a new system called PedroPathing. One highlight of this program is its ability to self-correct when off course, improving accuracy and effectiveness.

Lastly, for our build, we wanted to maximize our scoring output. From our past competitions, we noticed our horizontal and vertical Viper slides, were much too heavy, reducing speed and efficiency. To solve this, we switched to lighter Misumi slides. We designed custom mounting brackets, a custom belt-driven system, and a custom mounting point for our horizontal slide. For the horizontal slide, we also replaced the belt-driven extension with a lever system to improve speed, ensuring it matched the upgraded performance of our slides. These changes helped us create a faster, more precise robot.

TEAM COLLABORATION

We are always looking to collaborate with our other teams to make our robots the best they can be. This is done through coding, design and strategy to make everything as optimized as it can be. The calaborations are not just to improve our robot capering to our competitors but to show our very best result at every competition and improve team moral.

HOW WE INCOMPERATED TEAM COLLABORATIONS

For provincials, we knew that to be competitive we needed the option to be able to hang specimens. The reason for this is to have versatility with other teams, which would cause us to not be restricted on what we can do. This would ensure maximum points due to the fact that both teams could use the scoring method they are most comfortable with.

With our autonomous we believe that it is important to collaborate with all three teams when troubleshooting through errors that we run into. This will assure sucess for all 3 teams and give people the chance to learn from their mistakes

In conclusion, we believe that team callaboration is fundamental for all of our teams sucess and learning, but also understanding that when teams are helping each other out it is for that team to learn and grow. For example our senior team helping out our grade 11 team to help them with the barries of their autonomous that they encouter. In the end all of this leads to an enviroment where everyone is always learning and inovating which is a key stepping stone to our future where we have to collaborate in a workplace.